

2001

This Page for Writing Answers

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This image shows a blank white page. There are some faint, dark specks scattered across the surface, which appear to be scanning artifacts or dust particles. The overall texture is slightly grainy, typical of a scanned document page.

DO Not Use This Page for Writing Answers

T/

A/

Substitutional impurity defects.

In substitutional impurity defects the parent atom is substituted with a foreign atom. This will disturb the crystal making it have impurities.

0000000000
0000000000
0000000000
0000000000

Substitutional

Interstitial impurity defect.

In interstitial impurity defect the atom in the crystal is misplaced and moved to an interstitial space within the crystal.

Interstitial

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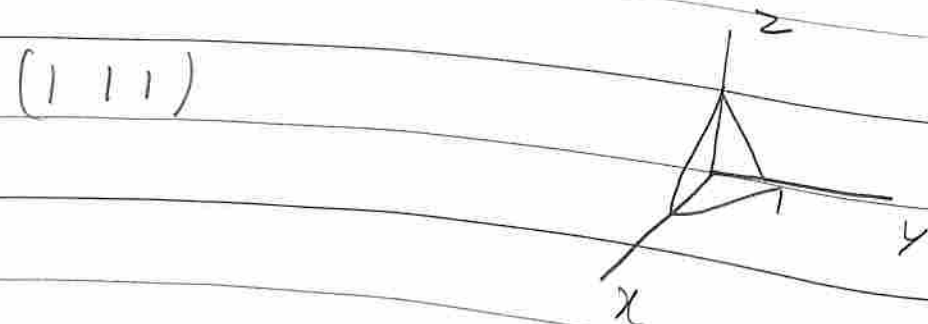
B

Edge dislocation	Screw dislocation
• This is line defect	• This is line defect
• Burgers vector $\perp$ dislocation	• Burgers vector $\parallel$ dislocation
• line of atoms stop	• line of atoms shifted
• stress producing	in axis.

C Miller indices. (MI)

Miller indices are the indices formed by miller to denote a crystal structure.

Any crystal structure can be denoted by miller indices and it is very helpful. It is in its simplest form.



D Laser.

Light amplification by Stimulated emission of rays.

Medical applications.

- Dentistry $\rightarrow$ can be used in dentistry to have a laser map out of the teeth.
- Surgery $\rightarrow$ It can be used in surgery for effortless and proper slicing and healing of the tissue. It is also precise.

Industry applications.

- Heavy metal cutting $\rightarrow$ since Lasers are highly intense and precise it can be used to cut metals with the same accuracy and precision on a larger scale. Human labour will also be reduced a lot.

E1 Laser.

Incandescent bulb.

Intensity

Intensity

It is focused on a
single point

it is not focused and more
everywhere

Coherence

Coherence

The photons move uniformly
in straight line

The photons don't move
uniformly and more randomly

Light emission

Light emission

The light is emitted
on a focus point,
single direction

The light is emitted
all the directions

F1 Optical fiber.

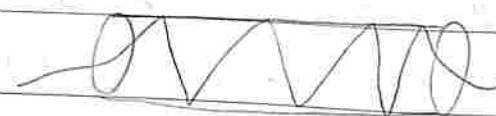
* Single index mode

* Multi index mode

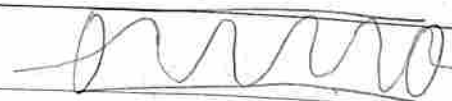
↳ Step index

↳ Graded index

In step index the waves will pass in a stepped format.



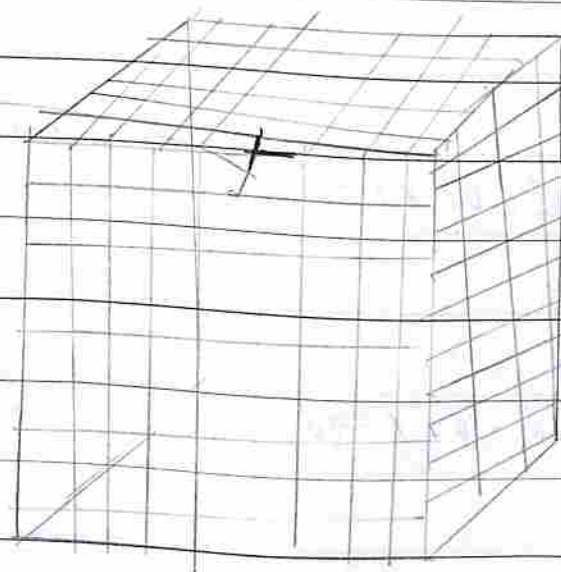
In graded index the waves will pass in a curved format



2

A1 Screw dislocation.

dislocation



This is one of the details of line dislocation.

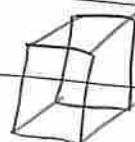
If an atom is dislocated and placed in another location of structure rather than its actual location then it is called dislocation.

In screw dislocation the Burgers vector is ~~perpendicular~~ ^{parallel} to the axis of dislocation. It may seem normal but there is dislocation.

B] Crystal Systems.

Cubic

$$a = b = c \quad \alpha = \beta = \gamma = 90^\circ$$



Rhombic

$$a = b \neq c \quad \alpha = \beta = \gamma = 90^\circ$$



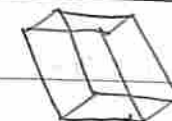
Orthorhombic

$$a \neq b \neq c \quad \alpha = \beta = \gamma = 90^\circ$$



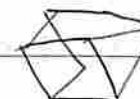
Monoclinic

$$a \neq b \neq c \quad \alpha = \beta = 90^\circ \neq \gamma$$



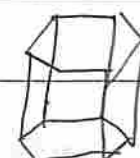
Triclinic

$$a \neq b \neq c \quad \alpha \neq \beta \neq \gamma$$



Hexagonal

$$a \neq b \neq c \quad \alpha = \beta = 90^\circ, \gamma = 120^\circ$$



C] Spontaneous

Stimulated

This is spontaneous emission

This is stimulated emission

moving from lower energy level to lower energy level without any help

moving from higher energy level to lower energy level with outside photon help.

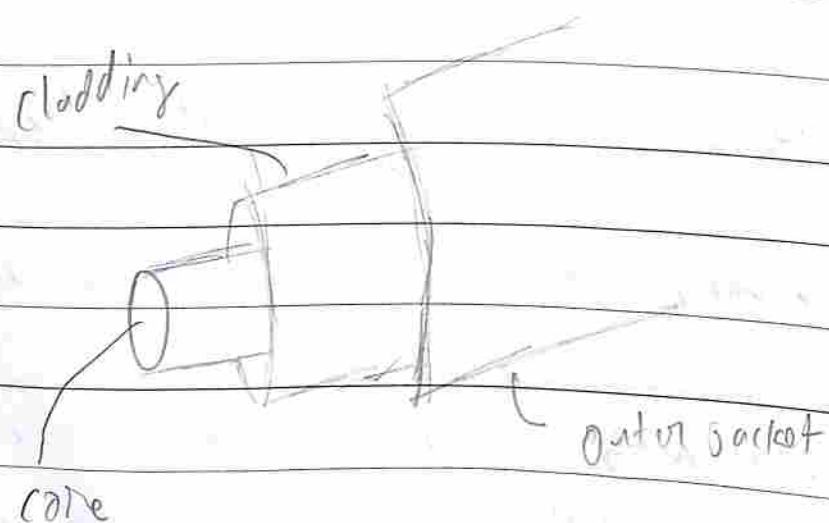
D Optical fibers

Single mode.

Only a single ^{beam} ray of light can pass through the optical fiber.

Multi mode

multiple beams or rays of light can pass through the optical fiber.



3/

1) Packing fraction.

$$PF = \frac{\text{vol of lattice}}{\text{vol of cube}} = \frac{\frac{4}{3}\pi^3}{a^3} \times Z$$

BCC SC

$$Z = 2 \quad \pi = a/2$$

$$= \frac{\frac{4}{3}\pi^3}{a^3} \times 2 = \frac{\frac{4}{3}(a/2)^3}{a^3} = \frac{\frac{4}{3}(a^3/8)}{a^3}$$

$$= \frac{4a^3}{3(8)a^3} = \frac{4}{3(8)} = \frac{1}{6} = 52\%$$

FCC BCC

$$Z = 4 \quad \pi = \sqrt{3}a$$

$$= \frac{\frac{4}{3}\pi^3}{a^3} \times 4 = \frac{\frac{4}{3}(\sqrt{3}a)^3}{a^3} = \frac{4 \times 3\sqrt{3}a^3}{a^3}$$

$$= 4\sqrt{3} \times 2 = 8\sqrt{3} = 64\%$$

FCC.

$$Z = 4 \quad R = a/\sqrt{2}$$

$$\frac{4/3 \pi r^3 \times 4}{a^3} = \frac{4/3 (\frac{a}{\sqrt{2}})^3}{a^3} = \frac{4 \cdot 95.}{3 \cdot 98. (2\sqrt{3})^3}$$

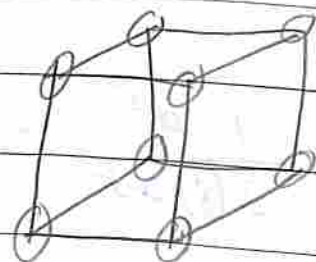
$$= \frac{4}{3(2\sqrt{2})} = \frac{4}{12\sqrt{2}} \quad \rightarrow \text{or } 6.8\%$$

B1

Lattice points

SC

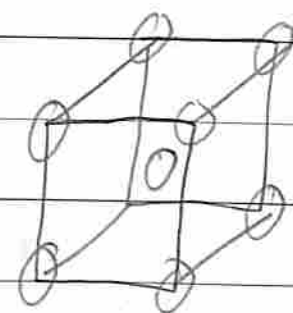
There are 1 point at each 8 corners
Each contribute $1/8$



$$\therefore 8 \times (1/8) = 1 \quad \therefore 1 \text{ Lattice point}$$

BCC

There are 1 in each corner
Each corner contributes $1/8$
There is 1 in the center

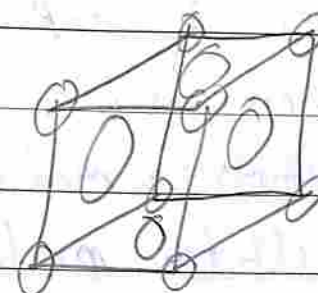


$$8 \times (1/8) + 1 = 1 + 1 = 2.$$

$\therefore 2$ lattice points

FCC

There are 1 in each corner
Each contribute $1/8$
There are 1 in each face
Each contribute $1/2$



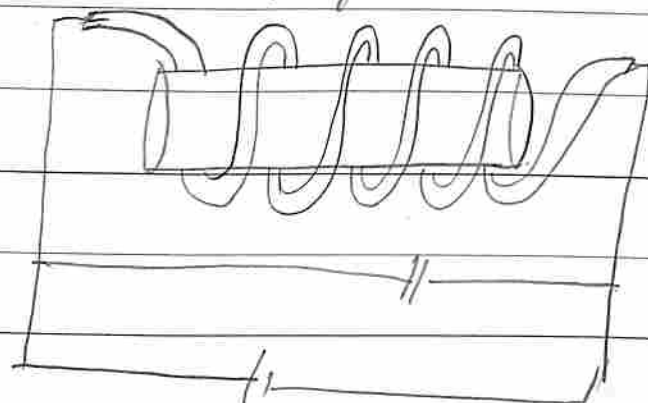
$$8 \times (1/8) + 6 \times (1/2) = 1 + 3 = 4$$

$\therefore 4$ Lattice points

5]

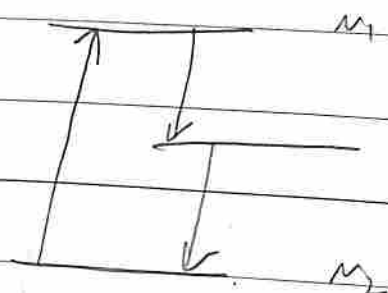
A]

Ruby laser



There are 3 components,

- active medium
- emission medium
- reflective panels.



The emission medium make the active medium active and makes the photons excited.

Then they get bounced in between the reflective panels back and forth. one of the side will be semi reflective panel and excited photon will pass through it then reveals to be laser.

B]

The components involved in general laser is

- Active medium
- emission medium
- reflective panels.
- The emission medium makes the active medium active and makes the photons excited.

Then they get bounced in between the reflective panels back and forth. one side will be semi reflective panel and excited photon will pass through it then reveals to be laser.

